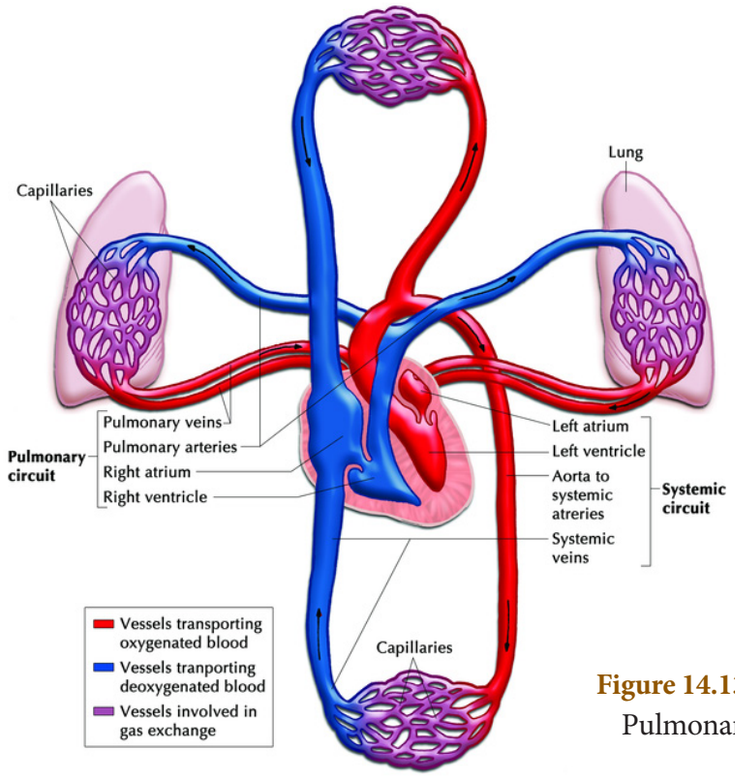


Figure 14.13 Systemic and Pulmonary circulation

14.15.2 Heart Beat

One complete **contraction** (**systole**) and **relaxation** (**diastole**) of the atrium and ventricles of the heart constitute heartbeat. The heart normally beats 72 – 75 times per minute.

More to Know

Neurogenic and Myogenic Heart Beat

Neurogenic heart beat is initiated by a nerve impulse caused from a nerve ganglion situated near the heart. e.g. Annelids, most arthropods

Myogenic heart beat is initiated by a specialized group of modified heart muscle fibres. e.g. Mollusca and Vertebrates

Initiation and conduction of Heart beat

The human heart is **myogenic** in nature. Contraction is initiated by a specialized portion of the heart muscle, the **sino-atrial** (SA) **node** which is situated in the wall of the right atrium near the opening of the **superior vena cava**. The SA node is broader at the top and tapering below. It is made up of thin fibres.

Sino-atrial node acts as the **'pacemaker'** of the heart because it is capable of initiating impulse which can stimulate the heart muscles to contract. The impulse from the sinoatrial node spreads as a wave of contraction over the right and left atrial wall pushing the blood through the **atrioventricular** valves into the ventricles. The wave of contraction from SA node reaches the **atrioventricular** (AV) **node** which is stimulated to emit an impulse of contraction spreading to the ventricular muscle via the **atrioventricular bundle** and **the Purkinje fibres**.



Atrioventricular bundle was discovered by His (1893). So is called Bundle of His.

Pulse: When the heart beats the blood is forced into the arteries. The expansion of the artery every time the blood is forced into it is called pulse. It can be felt by placing the fingertip on the artery near the wrist. Normal pulse rate ranges from 70 – 90 / min.

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